



Reactivity of mixed organozinc and mixed organocopper reagents: 12. Three component reaction of mixed (n-alkyl)(diaryl)zincates, chloroformates and phosphines for the synthesis of esters



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ABSTRACT

The reaction of mixed *n*-butyldiphenylzincate, *n*-BuPh₂ZnMgBr with ethyl chloroformate, ClCOOEt in the presence *n*-Bu₃P in THF takes place with quantitative yield and phenyl group transfer to give PhCOOEt. Ethoxycarbonylation of *n*-BuPh₂ZnMgBr is preferable to the reaction of PhMgBr forming ester and triphenylcarbinol and also to the reaction of triphenylzincate, Ph₃ZnMgBr for atom economy. Group selectivity in the phosphine catalyzed C–COOR coupling of *n*-BuPh₂ZnMgBr and *n*-Bu₂PhZnMgBr can be controlled by changing reaction parameters. *n*-Bu₃P catalyzed reaction of *n*-BuPh₂ZnMgBr with ClCOOEt takes place with phenyl selectivity whereas reaction of *n*-Bu₂PhZnMgBr with ClCOOPh results in *n*-butyl transfer. Catalysis by Ph₃P increases *n*-butyl group:phenyl group transfer ratio in the ethoxycarbonylation of both zincates. Selective transfer of aryl groups in *n*-Bu₃P catalyzed reaction of *n*-butyl(aryl)₂ZnMgBr reagents with ClCOOEt in THF provides a new procedure for the organometallic synthesis of arene-carboxylic acid ethyl esters at room temperature.

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1. Introduction

Organozinc reagents [1–3] are widely used organometallic reagents in organic synthesis due to their increased reactivity in the presence of transition metal and/or organic catalysis, availability of a number of methods for their preparation and their compatibility with many functional groups compared with organolithium and Grignard reagents.

Triorganozincates, R₃ZnM (M = Li, MgX; X = Br, Cl) represent a class of organozinc reagents which are more reactive than diorganozincates, R₂Zn and organylzinc halides, RZnX. However, triorganozincates have not been used in organic synthesis [4–8] as extensively as organozinc halides and diorganozincates.

The serious drawback in the synthetic use of R₂Zn and R₃ZnM reagents is the transfer of only one of the R groups to the electrophile leading to the low atom-economy in their reactions. To avoid this problem, mixed diorganozincates R¹R²Zn [9–12] and mixed triorganozincates R¹R²R³ZnM [13,14] have been developed, in which one of the R groups has a higher transfer rate than the other one. Mixed diorganozincates, R_RR_TZn [15–18] and triorganozincates,

R_T(R_R)₂ZnM [19–25] are synthetically useful reagents.

In the group selectivity of mixed cuprates R¹R²CuLi, the commonly accepted hypothesis is that the group in R¹R²CuLi that has a stronger bond to Cu acts as the group of lower selectivity [1,26–28].

In our serial work on the reactivity and group selectivity control of mixed organozinc and copper reagents [29–39], we observed that the group transfer of an organyl group as R_T in mixed diorganozincates R¹R²Zn [29–31,34,35], diorganocuprates, R¹R²CuM (M = MgBr [32,37,38], ZnCl [38]) and copper catalyzed mixed triorganozincates, R¹R²ZnMgBr [33,37] can be controlled by changing reaction parameters, i.e. counter ion, M, leaving group of the electrophile, solvent, temperature, transition metal catalyst and organic catalyst. We hypothesized that in reactions of mixed reagents, R¹R²Zn and R¹R²CuM, R¹ group can change not only the transfer ability, but also the transfer rate of R² group leading to different reaction rate of mixed reagents than that of homo reagents R₂Zn and R₂CuM [33,36,39]. We also developed new atom-economic procedures for C-organyl [32–34,37], C–N [31] and C–COR [29,30,34] coupling reactions using mixed diorganozinc reagents, R_RR_TZn. However, to the best of our knowledge, the reactivity and group selectivity in reactions of mixed organozincates R¹R²ZnM (M = Li, MgBr) has not been intensively investigated.

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Ester functionality is one of the most common functional groups and can be prepared using a number of metal free and metal catalyzed methods [40,41]. However, to the best of our knowledge, the use of C–COOR coupling of organolithium, -magnesium and -zinc reagents and diorganocuprates is very limited [42,43]. Due to the importance of a selective acylation procedure for carbanions, we focused our interest on the applicability of mixed R^1R^2Zn and $R^1(R^2)_2ZnMgBr$ reagents for organometallic synthesis of esters.

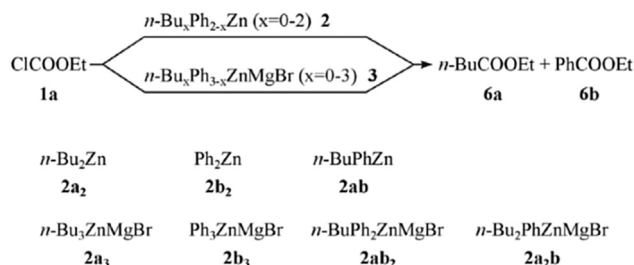
Here, we report our progress on the control of group selectivity of (*n*-alkyl)(aryl)₂zincates in acylation with chloroformate esters, ClCOOR and achieving a protocol for the conversion of $C_{aryl}-Zn$ bond to $C_{aryl}-COOR$ bond.

2. Results and discussion

As a model reaction to investigate the group selectivity of mixed R^1R^2Zn (**2**) and $R^1(R^2)_2ZnMgBr$ (**3**) reagents with chloroformates, we chose *n*-butyl and phenyl groups in these reagents to transfer to ethyl chloroformate, ClCOOEt (**1a**) in THF (Scheme 1). We also investigated the reaction of homo diorganozincs and homo triorganozincates to compare with mixed reagents for atom-economy in the synthetic application.

We used magnesium-based diorganozincs and triorganozincates. For the preparation of mixed diorganozinc **2ab**, PhZnCl was allowed to react with *n*-BuMgBr in a one pot procedure [30]. The same procedure can be also applied by reacting *n*-BuZnCl with PhMgBr. Mixed triorganozincates **3ab₂** and **3a₂b** were also prepared easily using *n*-butyl- and phenylzinc reagents and *n*-butyl- and phenyl Grignard reagents [34].

The original attachment of *n*-Bu or Ph groups to Zn or Mg does not make a change on the transfer ability of these groups in mixed diorganozincs and triorganozincates [34].



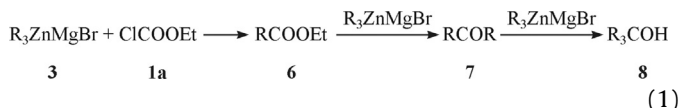
Scheme 1. Ethoxycarbonylation of homo and mixed diorganozincs, $n-Bu_xPh_{2-x}Zn$ ($x = 0-2$) (**2**) and mixed triorganozincates, $n-Bu_xPh_{3-x}ZnMgBr$ ($x = 0-3$) (**3**).

First, we determined the effect of the reaction parameters on the group selectivity of mixed triorganozincates, *n*-BuPh₂ZnMgBr (**3ab₂**) and *n*-Bu₂PhZnMgBr (**3a₂b**) in their reaction with ethyl chloroformate (**1a**) in THF. In the initial study, we determined the background yields using homozincates, trin-*n*-butylzincate (**3a₃**) and triphenylzincate (**3b₃**).

Table 1 summarizes the yields and group selectivities, i.e. *n*-Bu group: Ph group transfer ratios in the ethoxycarbonylation of homo zincates, **3a₃** and **3b₃** and mixed zincates, **3ab₂** and **3a₂b**. Transfer abilities of organyl groups were found by calculating the GC yields of ketones using authentic samples of **6a** and **6b**. The reaction temperature and time was optimized to be 25 °C and 1 h, respectively. The reaction of **3a₃** gave the product ester **6a** in 43% yield whereas **3b₃** afforded the ester **6b** in a lower yield of 22% (entries 1 and 2).

Ethoxycarbonylation of mixed zincate **3ab₂** resulted in a good total yield of 76% with a **6a:6b** ratio of 10:90 (entry 3). For the reaction of mixed zincate **3a₂b**, a total yield of 48% was obtained with a **6a:6b** ratio of 25:75 (entry 4).

The inherent difficulty in the reaction of organozincate reagents **3** with chloroformate ester **1a** is possible formation of ketones **7** and tert-alcohols **8** as side products (1). However, in the reaction of homo and mixed zincates, ketones were not detected and the yield of tert-alcohols, *n*-Bu₃COH and Ph₃COH did not exceed 10%.



As seen, much lower ethoxycarbonylation yields of *n*-Bu group, i.e. 8% and 12% were obtained using mixed zincates **3ab₂** and **3a₂b** respectively, compared to the yield obtained in the reaction of homozincate **3a₃**, i.e. 43%. However, we were excited by the ethoxycarbonylation yields of Ph group, i.e. 68% and 36% using mixed zincates **3ab₂** and **3a₂b**, respectively, which are higher than the yield obtained in the reaction of homozincate **3b₃**, i.e. 22%. Next, expecting a change in *n*-Bu group or Ph group transfer, we investigated the effect of transition metal catalysis and organic catalysis on the group selectivities in the ethoxycarbonylation of **3ab₂**. As transition metal catalysts, a range of Cu, Ag, Ni and Pd salts were screened (Table 1). The use of CuI as a catalyst lowered the total yield to 24% in the reaction of **3ab₂** (entry 5) and the yield was further decreased to 12% in the presence of CuCN and CuBr.Me₂S. The use of AgI or AgOCOCF₃ as a catalyst in the reaction of **3ab₂** led to a modest yield of about 40% (entries 6 and 7) with phenyl selective coupling. NiCl₂(PPh₃)₂ catalysis did not give a yield higher than 18% (entry 8). Ethoxycarbonylation in the presence of PdCl₂ did not proceed to completion with a yield of 44% (entry 9) and catalysis with Pd(OAc)₂ or PEPPSI-*i*Pr resulted in inferior yields of 36% and 4%, respectively with phenyl selectivity. Unreacted ethyl chloroformate **1a** was mostly recovered in transition metal-free and catalyzed reactions of zincates.

The reaction of homodiorganozincs, di-*n*-butylzinc (**2a₂**), diphenylzinc (**2b₂**) and mixed *n*-butylphenylzinc (**2ab**) in ethoxycarbonylation with **1a** did not take place and gave dramatically low yields of 7–10% in the presence of CuI catalysis.

In studying the effects of organic catalysis on the group selectivity of mixed triorganozincates in their reaction with **1a**, we used donor solvents as cosolvents in THF and Lewis bases as additives. In ethoxycarbonylation of **3ab₂** in THF, we used donor solvents, NMP, diglyme, DMF and also TMEDA as cosolvents. The results showed that increasing polarity of the solvent prevented the organyl groups from transferring to C=O group resulting in yields lower than 5%. The recovery of ethyl chloroformate **1a** only in 20–40% yield suggests a possible reaction between these N-donor and O-donor solvents and **1a**. In fact, it is known that tert-amines react with chloroformate esters to give carbamates under appropriate conditions [44]. However, we did not turn our attention to a possible reaction of donor solvents with **1a**.

We already found *n*-Bu₃P and Ph₃P as successful additives to carry out the acylation of *n*-BuPhZn (**2ab**) in high yield with complete *n*-Bu group transfer and developed a phosphine catalyzed acylation of mixed (*n*-alkyl)phenylzincs with aromatic acyl chlorides for the synthesis of *n*-alkylarylketones [30]. In acylation of Grignard reagents [45] and diorganozincs [30] possible *in situ* formation of acylphosphonium salts, (RCO)P⁺R₃Cl[−] have been reported [30,45]. So expecting formation of a similar phosphonium salt, (EtOOC)P⁺R₃Cl[−] from ClCOOEt (**1a**) and R₃P, we tried phosphine catalysis to see if the yield increases and/or group selectivity changes in ethoxycarbonylation (Table 2). We observed that *n*-Bu₃P catalysis proceeded the reaction with a quantitative yield and complete phenyl selectivity (entries 1 and 2) compared with

Table 1

Optimization on the yield and organic group selectivity in the ethoxycarbonylation of *n*-butyl and phenyl Grignard reagents derived homo and mixed triorganozincates **3** with **1a** in THF. Screening Cu, Ag, Ni and Pd catalysts.^{a,b}

$$\begin{array}{c}
 n\text{-Bu}_x\text{Ph}_{3-x}\text{ZnMgBr} \text{ (} x=0\text{--}3\text{)} + \text{ClCOOEt} \xrightarrow[\text{THF, 25}^\circ\text{C, 2h}]{\text{Catalyst (5 mol\%)}} n\text{-BuCOOEt} + \text{PhCOOEt} \\
 \mathbf{3} \qquad \qquad \mathbf{1a} \qquad \qquad \qquad \mathbf{6a} \qquad \qquad \mathbf{6b} \\
 \\
 \begin{array}{cccc}
 n\text{-Bu}_3\text{ZnMgBr} & \text{Ph}_3\text{ZnMgBr} & n\text{-BuPh}_2\text{ZnMgBr} & n\text{-Bu}_2\text{PhZnMgBr} \\
 \mathbf{3a_3} & \mathbf{3b_3} & \mathbf{3ab_2} & \mathbf{3a_2b}
 \end{array}
 \end{array}$$

Entry	Zincate	Catalyst	Yield, % ^c			Group selectivity ^e 6a:6b
			6a	6b	Total ^d	
1	3a₃	—	43			—
2	3b₃	—		22		—
3	3ab₂	—			76	10:90
4	3a₂b	—			48	25:75
5	3ab₂	CuI			24	12:88
6	3ab₂	AgI			41	7:93
7	3ab₂	Ag(OCOCF ₃)			43	9:91
8	3ab₂	NiCl ₂ (PPh ₃) ₂			8	0:100
9	3ab₂	PdCl ₂			44	9:91

^a The reactions were carried out according to the conditions indicated by the above equation. Molar ratio of **3:1a** is 2:1.

^b All the data are average of five or six experiments.

^c GC yield of **6a** in the reaction of **3a₃** and **2a₂** and GC yield of **6b** in the reaction of **3b₃**.

^d The sum of GC yields of **6a** and **6b** in the reactions of **3ab₂** and **3a₂b**.

^e The ratio of GC yields of **6a** and **6b**.

uncatalyzed reaction (Table 1, entry 3). Optimal amount of *n*-Bu₃P catalyst was found to be 10 mol%. However, when 10 mol% of Ph₃P was used, both yield and phenyl selectivity decreased (entry 3). The use of Ph₃P in 25–50 mol% gave still inferior results on the yield and phenyl selectivity (entries 4 and 5) compared to the outcome of the reaction in the presence of *n*-Bu₃P catalysis.

These results showed that *n*-Bu₃P is superior to Ph₃P in catalysis for aryl selective ethoxycarbonylation of (*n*-alkyl)(aryl)₂ZnMgBr reagents with ClCOOEt **1a**.

Mixed zincate **3a₂b** behaved similarly either in the presence of 10 mol% *n*-Bu₃P or 50 mol % Ph₃P catalysis and ethoxycarbonylation yield increased (entries 6 and 7); however phenyl selectivity decreased in the presence of *n*-Bu₃P (entry 7) compared with uncatalyzed reaction (Table 1, entry 4). For comparison with the quantitative and phenyl selective *n*-Bu₃P catalyzed ethoxycarbonylation of *n*-BuPh₂ZnMgBr (**3ab₂**), we also carried out the reaction with Ph₃ZnMgBr (**3b₃**) and found 90% yield (entry 8). This result clearly shows the atom-economic character of **3ab₂** for Ph

Table 2

Effect of N- and O-donor cosolvents and Lewis base additives on the yield and group selectivity in the ethoxycarbonylation of Grignard reagents derived triphenylzincate **3b₃**, *n*-butyldiphenylzincate **3ab₂** and di-*n*-butylphenylzincate **3a₂b** with **1a** in THF.^{a,b}

$$\begin{array}{c}
 n\text{-Bu}_x\text{Ph}_{3-x}\text{ZnMgBr} + \text{ClCOOEt} \xrightarrow[\text{THF : cosolvent, 25}^\circ\text{C, 2h}]{\text{Additive}} n\text{-BuCOOEt} + \text{PhCOOEt} \\
 \mathbf{3} \qquad \qquad \mathbf{1a} \qquad \qquad \qquad \mathbf{6a} \qquad \qquad \mathbf{6b} \\
 \\
 \begin{array}{ccc}
 \text{Ph}_3\text{ZnMgBr} & n\text{-BuPh}_2\text{ZnMgBr} & n\text{-Bu}_2\text{PhZnMgBr} \\
 \mathbf{3b_3} & \mathbf{3ab_2} & \mathbf{3a_2b}
 \end{array}
 \end{array}$$

(*x*=0–2)

Entry	Zincate	Solvent ^c /Additive	Total yield, % ^d	Group selectivity ^e 6a:6b
1	3ab₂	THF/ <i>n</i> -Bu ₃ P (10 mol%)	98	4:96
2	3ab₂	THF/ <i>n</i> -Bu ₃ P (25 mol%)	100	5:95
3	3ab₂	THF/Ph ₃ P (10 mol%)	47	18:82
4	3ab₂	THF/Ph ₃ P (25 mol%)	61	39:61
5	3ab₂	THF/Ph ₃ P (50 mol%)	74	43:57
6	3a₂b	THF/ <i>n</i> -Bu ₃ P (10 mol%)	73	27:73
7	3a₂b	THF/Ph ₃ P (50 mol%)	67	43:57
8	3b₃	THF/ <i>n</i> -Bu ₃ P (25 mol%)	90	—

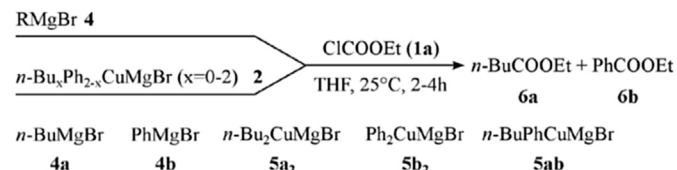
^a The reactions were carried out according to the conditions indicated by the above equation. Molar ratio of **3:1a** is 2:1.

^b All the data are average of five or six experiments.

^c THF:cosolvent ratio was optimized to be 2:1.

^d The sum of GC yields of **6a** and **6b** in the reactions of **3ab₂** and **3a₂b**. GC yield of **6b** in the reaction of **3a₃**.

^e The ratio of GC yields of **6a** and **6b**.



Scheme 2. Ethoxycarbonylation of RMgBr ($R = n\text{-Bu, Ph}$) reagents (**4**) and homo and mixed diorganocuprates, $n\text{-Bu}_x\text{Ph}_{2-x}\text{CuMgBr}$ ($x = 0-2$) (**5**).

group transfer in ethoxycarbonylation.

We also examined the reactivity of n -butyl and phenyl Grignard reagents (**4**) and Grignard reagent derived diorganocuprates (**5**) in ethoxycarbonylation with **1a** aiming a comparison of their yield and/or group selectivity with those of triorganozincates (**3**) (Scheme 2). The difficulty is that formation of ketones and/or tert-alcohols as side products appeared in the reaction of Grignard reagents and a product mixture of ester **6**, ketone **7** and mostly tert-alcohol **7** was obtained.

The results are summarized below:

The reaction of $n\text{-BuMgBr}$ (**4a**) with **1a** in THF resulted in 50% yield of the product mixture containing **6a**, $n\text{-Bu}_2\text{CO}$ and mostly $n\text{-Bu}_3\text{COH}$. The use of 10 mol% of CuI as a catalyst decreased the yield of product mixture to 26%. R_3P ($R = n\text{-Bu, Ph}$) catalyzed reaction led to 55% yield of product mixture. In the reaction of PhMgBr (**4b**) with **1a**, we found 55% of product mixture containing **6b**, Ph_2CO and mostly Ph_3COH . However, catalytic reaction with 10 mol % CuI increased the yield to 70%, in sharp contrast to that observed in the copper catalyzed reaction of $n\text{-BuMgBr}$ (**4a**). $n\text{-Bu}_3\text{P}$ catalysis also led to 70% yield of product mixture.

Treatment of **1a** with pre-prepared di- n -butylcuprate, $n\text{-Bu}_2\text{CuMgBr}$ (**5a2**) and diphenylcuprate, Ph_2CuMgBr (**5b2**) resulted in quite low yields of ethoxycarbonylation, i.e. 17% yield of **6a** and 30% yield of **6b**, respectively. Phosphine catalysis did not lead to any improvement uncatalyzed and phosphine catalyzed ethoxycarbonylation of mixed cuprate, $n\text{-BuPhCuMgBr}$ (**5ab**) led to a yield of 14% with $n\text{-Bu}$ group selectivity.

In summary, it is obvious that for a clean and high yield $\text{C}_{\text{aryl}}\text{—COOEt}$ coupling, the use of Grignard reagent derived mixed n -alkyldiarylzincates is preferable aryl Grignard reagents and also to triarylzincates to avoid the waste use of an aryl group.

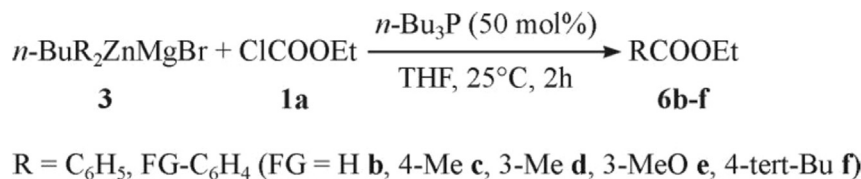
After having established the optimal reaction conditions, we explored the scope of the $n\text{-Bu}_3\text{P}$ catalyzed aryl selective ethoxycarbonylation of (n -butyl)(diaryl)zincates with ClCOOEt (**1a**) to use in the organometallic synthesis of esters. The results are summarized in Table 3. The problem was the isolation of the esters from the product mixture due to the presence of triarylcarbinols as side products although less than 10% yields. The chromatographic separation of an ester-alcohol mixture is known as a tedious and time consuming procedure and the use of DES (deep-eutectic solvents) have been reported for successful separation of esters from alcohols [46]. DES are a new type of ionic solvents formed by mixing a quaternary ammonium salt together with a hydrogen-bond-donor. We used choline chloride:glycerol (1:2 equiv/1 equiv) for extraction of product esters.

As seen, a range of aryl groups of ($n\text{-Bu}$)($\text{FG—C}_6\text{H}_4$) ZnMgBr reagents were transferred selectively to ClCOOEt (**1a**) to afford the desired products, $\text{FG—C}_6\text{H}_4\text{COOEt}$ in modest to good yields. However, ethoxycarbonylation of Br— and EtOOC— substituted phenyl groups failed to give the desired products in acceptable yields.

To further expand the application scope of the C—COOR coupling reactions of organozincates, we also moved our attention to the effect of the —OR group of XCOOR reagents, as a new parameter on the yield and group selectivity. For an initial study, we employed phenyl chloroformate, ClCOOPh (**1b**) in $n\text{-Bu}_3\text{P}$ catalyzed reactions. The yields and group selectivities in the phenoxycarbonylation of homo and mixed zincates are given in Table 4. Phosphine catalyzed reaction of zincates with **1b** also proceeded smoothly and did not go to completion after extended reaction time. It is quite interesting that **1b** enhanced n -butyl selectivity remarkably in the reaction with mixed zincates. Mixed zincate, $n\text{-BuPh}_2\text{ZnMgBr}$ (**3ab2**) gave 87% yield and **9a:9b** ratio of 54:46 in the reaction with ClCOOPh (**1b**) (entry 3) compared to 98% yield and **7a:7b** ratio of 4:96 in the reaction with ClCOOEt (Table 2, entry 1). Mixed zincate, $n\text{-Bu}_2\text{PhZnMgBr}$ (**3a2b**) also led to 87% yield with n -butyl selectivity (**9a:9b** = 90:10) in the reaction with ClCOOPh (**1b**) (entry 4) compared to 73% yield and **6a:6b** ratio of 27:73 in the reaction with ClCOOEt (**1a**) (Table 2, entry 6). Phenoxycarbonylation yield of $n\text{-Bu}$ group in homozincate, $n\text{-Bu}_3\text{ZnMgBr}$ (**3a3**) and Ph group in homozincate, Ph_3ZnMgBr (**3b3**) resulted in 42% and 76%, respectively (Table 4, entries 1 and 2). These experiments also showed the atom-economic character of the n -butyl selective reaction of $n\text{-Bu}_2\text{PhZnMgBr}$ (**3a2b**) with **1b** to give $n\text{-BuCOOPh}$.

We were pleased to find that in the presence of phosphine catalysis, ClCOOEt (**1a**) and ClCOOPh (**1b**) are workable

Table 3
Trin-butylphosphine catalyzed ethoxycarbonylation of bromomagnesium n -butyldiarylzincates with ethyl chloroformate (**1a**) in THF at 25 °C in 2 h.^a



Entry		Coupling yield, % ^{b,c}
1	C_6H_5 b	78
2	4-Me C_6H_4 c	81
3	3-Me C_6H_4 d	64
4	3-MeOC C_6H_4 e	58
5	4-tert-BuC C_6H_4 f	75

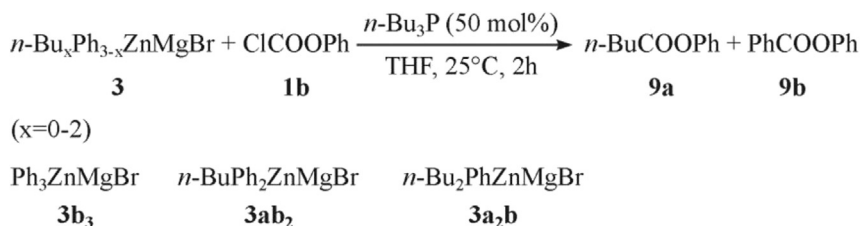
^a Reactions were run with a 2:1 M ratio of **3:1a**. For synthetic procedure, see Experimental Section.

^b Isolated yield. Average of at least two independent experiments.

^c The products were fully characterized by ^1H NMR.

Table 4

Trin-butylphosphine catalyzed phenoxycarbonylation of Grignard reagents derived triphenylzincate (**3b₃**), *n*-butyldiphenylzincate (**3ab₂**) and di-*n*-butylphenylzincate **3a₂b** with phenyl chloroformate **1b** in THF.^{a,b}



Entry	Zincate	Yield, % ^c			Group selectivity ^e 9a:9b
		7a	7b	Total ^d	
1	3a₃	42			—
2	3b₃		76		—
3	3ab₂			87	54:46
4	3a₂b			87	90:10

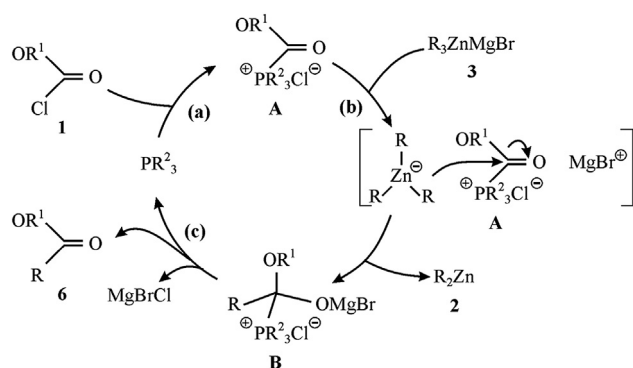
^a The reactions were carried out according to the conditions indicated by the above equation. Molar ratio of **3:1b** is 2:1.

^b All the data are average of five or six experiments.

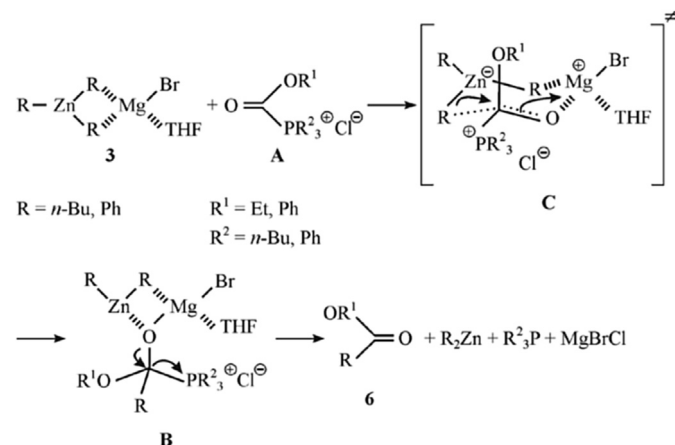
^c GC yield of **9a** in the reaction of **3a₃** and GC yield of **9b** in the reaction of **3b₃**.

^d The sum of GC yields of **9a** and **9b** in the reactions of **3ab₂** and **3a₂b**.

^e The ratio of GC yields of **9a** and **9b**.



Scheme 3. Proposed catalytic cycle for phosphine, R_3P catalyzed carbonylation of triorganozincates, R_3ZnMgBr with chloroformates, ClCOOR^1 .



Scheme 4. Catalytic intermediate **A** and transition state **C** in R_3P catalyzed reaction of R_3ZnMgBr with ClCOOR^1 .

substrates to react with mixed zincates, $n\text{-alkyl(aryl)}_2\text{ZnMgBr}$ and $(n\text{-alkyl})_2(\text{aryl})\text{ZnMgBr}$ in THF resulting in group selective transfer of *n*-alkyl or aryl groups. In the presence of *n*-Bu₃P,

n-alkyl(aryl)₂ZnMgBr reagents give C_{aryl}–COOEt coupling with **1a** and $(n\text{-alkyl})_2(\text{aryl})\text{ZnMgBr}$ reagents give C_{alkyl}–COOPh coupling in high yield at room temperature.

Depending on the catalytic formation of $(\text{ROOC})\text{P}^+\text{R}_3\text{Cl}^-$ ions as oxycarbonylation reagents, we offer a catalytic cycle for R_3P catalyzed reaction of triorganozincates, R_3ZnMgBr with ClCOOR^1 to give esters RCOOR^1 (Scheme 3).

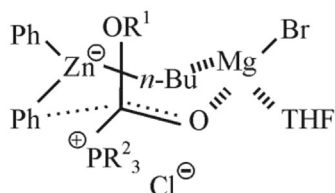
The structure of stoichiometric triorganozincates in solution has been reported as solvent separated ion pairs (SSIP) or as contact ion pairs (CIP) of Zn centered anions and counter cations $\text{R}^1\text{R}_2^2\text{Zn}^-\text{M}^+$ ($\text{R}^1 = \text{R}^2$ or $\text{R}^1 \neq \text{R}^2$, $\text{M} = \text{Li, MgBr}$) [1]. However, Nakamura et al. reported that triorganozincates react as organyl nucleophiles, but not as zinc centered nucleophiles [47]. Hevia et al. has revealed the formation of solvent separated R_3ZnMgBr structure, $[\text{R}_3\text{Zn}^-\text{Mg}_2\text{Cl}_2^+](\text{THF})_6$ in the solid state and in solution [8,48]. Ishihara et al. recently postulated ion pairs as essential moiety of zincate reagents in the reaction of homo and mixed zincates to ketones [49].

In the catalytic cycle, in step b, zincate anion R_3Zn^- moiety attacks the *in situ* formed catalytic phosphonium salt **A** yielding addition product **B** and R_2Zn . In step c, **B** releases the product ester and the catalyst to regenerate the phosphonium salt **A** in the following step a.

However, we observed that phosphine catalyzed reactions of mixed triorganozincates with haloformate esters takes place with group selectivity, which depends on the reaction parameters. In the *n*-Bu₃P catalyzed reaction of *n*-BuPh₂ZnMgBr (**3ab₂**) and *n*-Bu₂PhZnMgBr (**3a₂b**) with ClCOOEt (**1a**), complete and higher phenyl selectivity is observed, respectively, whereas their reaction with ClCOOPh (**1b**) results in higher and complete *n*-butyl selectivity, respectively. It seems that the reaction of haloformate ester with –OPh group increases *n*-Bu group:Ph group transfer ratio than that observed in the reaction of haloformate ester with –OEt group. In addition, Ph₃P catalysis increases *n*-Bu group:Ph group transfer ratio in the reaction of **3ab₂** and **3a₂b**.

For a plausible explanation for dependence of group selectivity of zincates **3ab₂** and **3a₂b** to leaving groups, –OR in ClCOOR ($\text{R} = \text{Et}$ or Ph) and also to phosphine catalysts, R_3P ($\text{R} = n\text{-Bu}$ or Ph), we assumed a six-membered ring chair conformation for coordination of R_3ZnMgBr to ClCOOR in the transition state **C** (Scheme 4) [16,50].

As magnesium prefers tetracoordination, we assume here the ester group will not participate in the chelation [29]. In the transition state **C**, $R^\ominus$ (n -Bu or Ph) will attack to the C=O group of the phosphonium salt **A** to afford the addition product **B** coordinated to R_2Zn . We think it is most probable that in transition states n -Bu group, being a smaller group will link Mg and Zn centers more easily than Ph group as reported for other methyl(aryl)₂zincates [51].



C for n -BuPh₂ZnMgBr₂ (**3ab₂**)

According to the model for **3ab₂** the attack of only Ph group to C=O group takes place yielding phenyl selectivity. For **3a₂b**, either n -Bu or Ph group will have the chance to react resulting in a decrease in phenyl coupling. In the case of the reaction with phenyl haloformate, ClCOOPh (**1b**), due to a possible steric interaction of $-OPh$ group in pseudoaxial or pseudoequatorial position, n -Bu group both in **3ab₂** and in **3a₂b** will have more chance to attack resulting in a higher n -Bu group:Ph group transfer ratio in **3ab₂** and n -butyl selectivity in **3a₂b**. This explanation seems to explain the observed differences in group selectivities of **3ab₂** and **3a₂b** in their reactions with haloformates, ClCOOR ($R = Et, Ph$). For the increase in n -Bu group:Ph group transfer ratio by using Ph_3P instead of n -Bu₃P, we simply think that the attack of Ph group in both **3ab₂** and **3a₂b** will not be easier due to the steric interaction of $-PR_3^+$ group in this case.

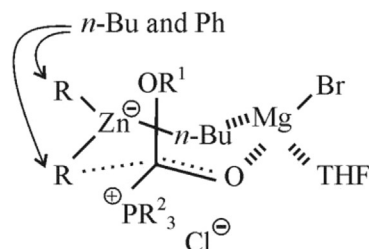
3. Conclusions

With the aim of controlling the group selectivity and providing a new method for $C_{aryl}-COOR$ coupling using mixed zincates, herein we have demonstrated that:

- Among the organomagnesium, -copper and -zinc reagents, reaction of mixed triorganozincates, $R^1R^2R^3ZnMgBr$ ($R^1, R^2 = n$ -alkyl or aryl) with ClCOOR ($R = Et, Ph$) in the presence of phosphines are successful for R^2-COOR coupling.
- In phosphine catalyzed reaction of $R_R(R_T)_2ZnMgBr$ with ClCOOR, group selectivity can be controlled by changing reaction parameters. n -Bu₃P catalyzed reaction of n -BuPh₂ZnMgBr with ClCOOEt takes place with high yield and phenyl group transfer whereas reaction of n -Bu₂PhZnMgBr with ClCOOPh leads to high yield with n -butyl group transfer. In addition, Ph_3P catalysis increases n -Bu group:Ph group transfer ratio in the reaction of both n -BuPh₂ZnMgBr n -Bu₂PhZnMgBr. These findings also provide support for our hypothesis that the group selectivity of mixed organozinc reagents depends on the reaction parameters.
- Selective transfer of aryl group in n -Bu₃P catalyzed reaction of $(n$ -butyl)(aryl)₂ZnMgBr reagents with ClCOOEt in

THF at room temperature is a newly developed procedure for the synthesis of esters using carbanions as nucleophiles.

- Further applications of mixed triorganozincate reagents for C-COOR and C-CONR₂ coupling and detailed mechanistic work for group selectivity are currently underway in our laboratories.



C for n -Bu₂PhZnMgBr₂ (**3a₂b**)

4. Experimental

4.1. General

All reactions were carried out under a nitrogen atmosphere in oven dried glassware using standard syringe-septum cap techniques [52]. Quantitative GC analysis were performed on a Thermo Finnigan gas chromatograph equipped with a 28-5 capillary column packed with phenylpolysiloxane using the internal standard technique. THF was distilled from sodium benzophenone dianion. Commercially available n -butyl bromide and aryl bromides were obtained commercially and purified using literature procedures. Grignard reagents were prepared in THF by standard methods. $ZnCl_2$ was dried under reduced pressure at 100 °C and used as THF solution.

n -Bu₃ZnMgBr (**1a₃**) and Ph_3 ZnMgBr (**1b₃**) were prepared by dropwise addition of 3 mol equiv of n -BuMgBr or $PhMgBr$, respectively in THF to a solution of 1 mol equiv of $ZnCl_2$ in THF at 15 °C and stirring at that temperature for 15 min. In the preparation of n -BuPh₂ZnMgBr (**1ab₂**) and n -Bu₂PhZnMgBr (**1a₂b**), freshly prepared Ph_2Zn and $PhZnCl$ were used, respectively. Ph_2Zn and $PhZnCl$ were prepared in THF by dropwise addition of 2 mol equiv or 1 mol equiv of $PhMgBr$, respectively in THF to a solution of 1 mol equiv of $ZnCl_2$ in THF at -15 °C and stirring at that temperature for 5 min. For the preparation of n -BuPh₂ZnMgBr (**1ab₂**) 1 mol equiv of n -BuMgBr was added to 1 mol equiv of Ph_2Zn in THF and for the preparation of n -Bu₂PhZnMgBr (**1a₂b**) 2 mol equiv of n -BuMgBr was added to $PhZnCl$ in THF, both at -15 °C and stirring at that temperature for 15 min.

4.2. Typical procedure for the synthesis of ethyl arenecarboxylates from bromomagnesium n -butyldiarylzincates and ethyl chloroformate in the presence of Ph_3P

In a two-necked flame-dried flask, mixed n -butyldiarylzinc bromomagnesium reagent, n -BuR₂ZnMgBr (**1ab₂** – **1af₂**) ($R = C_6H_5$ **b**, 4-MeC₆H₄ **c**, 3-MeC₆H₄ **d**, 4-MeOC₆H₄ **e**, 3-MeOC₆H₄) (10 mmol) was freshly prepared as explained above for **3ab₂**. For this purpose, the flask cooled to -15 °C was charged with $ZnCl_2$ (10 mmol) in THF (10 ml). Arylmagnesium bromide (20 mmol) was added at -15 °C. Following stirring for 15 min at that

temperature, *n*-BuMgBr (10 mmol) in THF was added at -15°C and heterogeneous solution of mixed zincate was stirred at that temperature for 15 min. To the mixed zincate (10 mmol), first *n*-Bu₃P (2.5 mmol) and after stirring for 5 min, ethyl chloroformate (**2a**) (5 mmol) was added dropwise. The reaction mixture was stirred at room temperature for 2 h. The mixture was hydrolyzed with H₂O. The separated organic phase was washed with 6NHCl to remove *n*-Bu₃P. Aqueous phase was extracted three times with Et₂O. The combined ethereal solutions were washed with aq. Na₂CO₃, dried over Na₂SO₄ and concentrated by rotary evaporation. The ethereal solution was mixed with DES (choline chloride:glycerol = 1:2 equiv/equiv) (sample: DES = 1:5 v/v) and centrifuged four times. The upper product ester phase was extracted with ethyl acetate and H₂O. The solutions were concentrated by rotary evaporation and subjected to silica gel column chromatography with petroleum ether: ether (1:1) as eluent to give product esters. Product esters were pure according to GC and ¹H NMR analysis. For the reusability of DES, H₂O was added to the lower ionic solvent phase and extracted with ethyl acetate to remove the product ester. Subsequently, H₂O was removed by heating the DES solution at 140°C in an open vessel for 2 h.

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